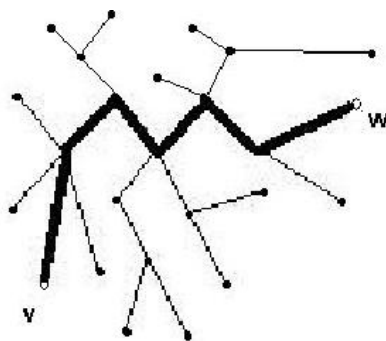


Discrete Mathematics

Lecture on Tree

Definition of Tree

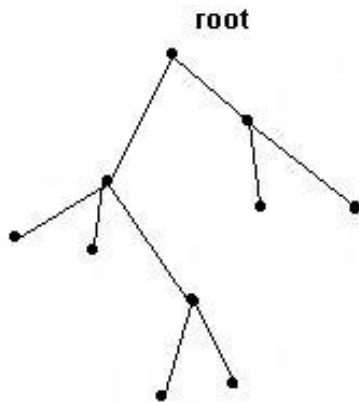


A *tree* T is

- simple graph such that for every pair of vertices v and w there is a unique path from v to w .

A tree is an acyclic graph or graph having no cycles.

Rooted tree



- A rooted tree is a tree where one of its vertices is designated the root else it is a free tree

Level of a vertex and tree height

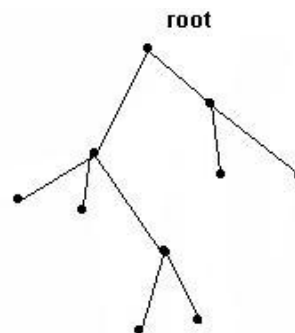
Let T be a rooted tree:

- The *level* $l(v)$ of a vertex v is the length of the simple path from v to the root of the tree
- The *height* h of a rooted tree T is the maximum of all level numbers of its vertices:

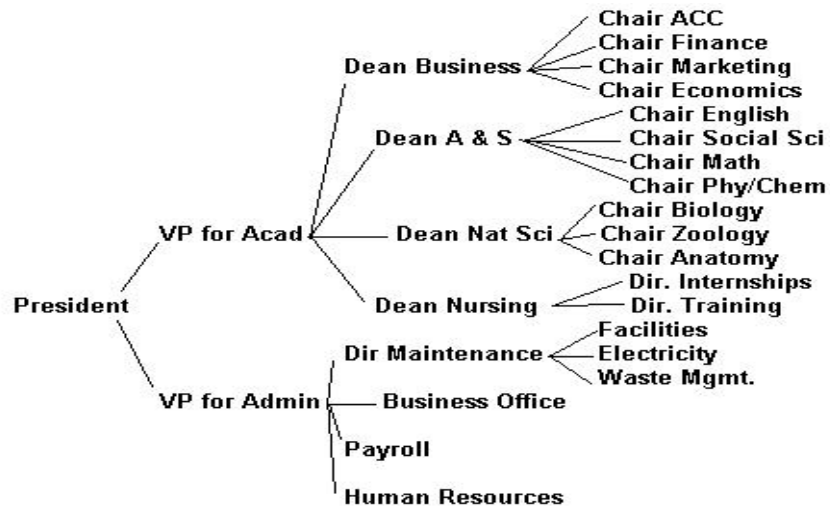
$$h = \max_{v \in V(T)} \{ l(v) \}$$

- Example:

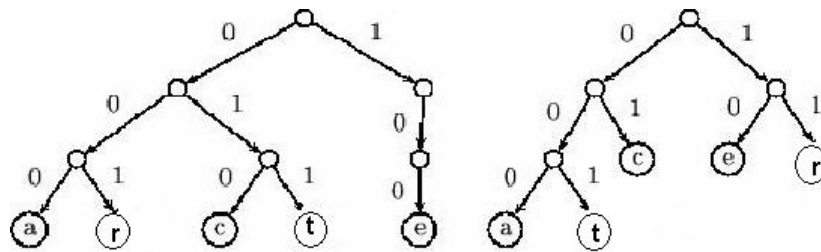
- the tree on the left (you see) has height 3



Organizational charts as Tree



Huffman codes



- On the left tree the word **rate** is encoded

001 000 011 100

- On the right tree, the same word **rate** is encoded

11 000 001 10

Given characters with frequencies

Combine the two smallest nodes

Pick the smallest frequencies:

$$A(5) + B(9) = 14$$

Repeat the process

New list:

C(12), D(13), (AB)(14), E(16), F(45)

$$C(12) + D(13) = 25$$

Continue combining

(AB)(14), E(16), (CD)(25), F(45)

$$\bullet (AB)(14) + E(16) = 30$$

Character	Frequency
A	5
B	9
C	12
D	13
E	16
F	45

Continue

(CD)(25), (ABE)(30), F(45)

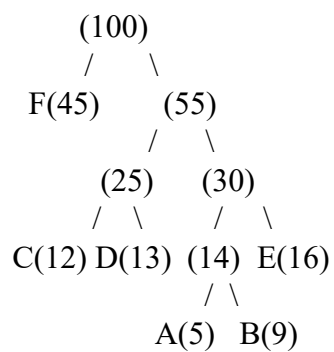
$$(CD)(25) + (ABE)(30) = 55$$

Final step

$$F(45) + (55) = 100$$

	Character	Frequency
(100)	A	5
/ \	B	9
F(45) (55)CDABE	C	12
/ \	D	13
CD(25) (30)ABE	E	16
/ \ / \	F	45
C(12) D(13) AB(14) E(16)		
/ \		
A(5) B(9)		

Assign codes (Left = 0, Right = 1)



Character	Code
F	0
C	100
D	101
A	1100
B	1101
E	111

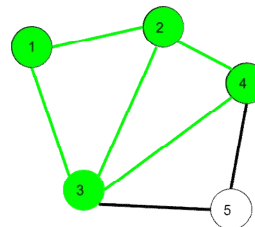
FACE, BAD, BED, FADE

0 1100 100 111,

Walk, Path, Circuit, Cycle

Walk : A walk can be defined as a sequence of edges and vertices of a graph. In a walk there can be repeated edges and vertices. In a graph there can be more than one walk.

Edges can be repeated
Vertex can be repeated



Open walk and Closed walk

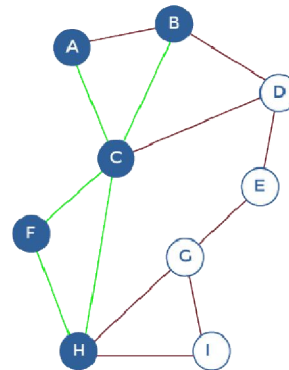
Open walk: A walk is said to be an open walk if the starting and ending vertices are different i.e. the origin vertex and terminal vertex are different.

Closed walk: A walk is said to be a closed walk if the starting and ending vertices are identical i.e. if a walk starts and ends at the same vertex, then it is said to be a closed walk.

Trails

A trail can be described as an open walk where no edge is allowed to repeat. In the trails, the vertex can be repeated.

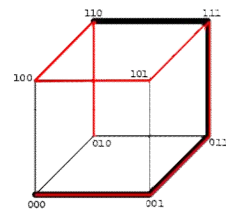
Tail = A, C, H, F, C, B



Path

A path is a type of open walk where neither edges nor vertices are allowed to repeat. There is a possibility that only the starting vertex and ending vertex are the same in a path.

Edges cannot be repeated
Vertex cannot be repeated

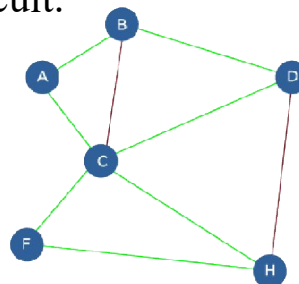


Hamiltonian path in red

Circuit

A circuit can be described as a closed walk where no edge is allowed to repeat. In the circuit, the vertex can be repeated. A closed trail in the graph theory is also known as a circuit.

Edges cannot be repeated
Vertex can be repeated

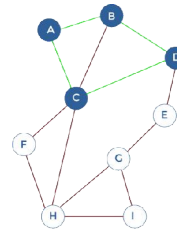


Circuit: A, B, D, C, F, H, C, A

Cycle

A closed path in the graph theory is also known as a Cycle. A cycle is a type of closed walk where neither edges nor vertices are allowed to repeat. There is a possibility that only the starting vertex and ending vertex are the same in a cycle.

Edges cannot be repeated
Vertex cannot be repeated



Cycle: A, B, D, C, A

Summary

Walk: Vertex may repeat, edges may repeat (closed or Open)

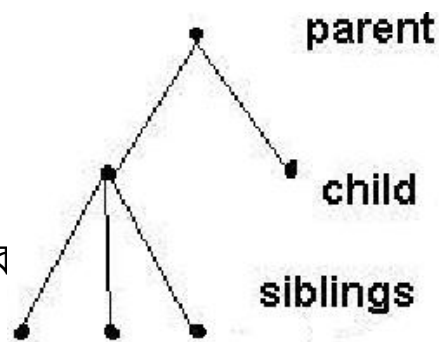
Circuit: Vertex may repeat, edges can not repeat (closed)

Cycle: Vertex can not repeat, edges can not repeat.

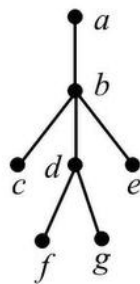
Path: Vertex can not repeat, edges can not repeat

7.2 Terminology

- Parent অভিভাবক
- Ancestor পূর্বপুরুষ
- Child সন্তান
- Descendant বংশধর
- Siblings সহোদর
- Terminal vertices প্রান্তিক শীর্ষবিন্দু
- Internal vertices অভ্যন্তরীণ শীর্ষবিন্দু
- Subtrees উপগাছ



Def:

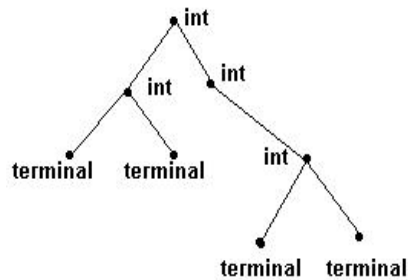


a is the **parent** of b , b is the **child** of a ,
 c, d, e are **siblings**,
 a, b, d are **ancestors** of f
 c, d, e, f, g are **descendants** of b
 c, e, f, g are **leaves** of the tree ($\deg=1$)
 a, b, d are **internal vertices** of the tree
 (at least one child)

subtree with d as its root:



Internal and external vertices

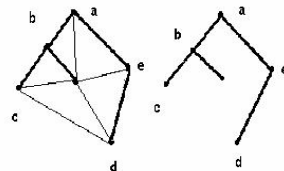


- An *internal vertex* is a vertex that has at least one child
- A *terminal vertex* is a vertex that has no children
- The tree in the example has 4 internal vertices and 4 terminal vertices

Spanning trees

Given a graph G , a tree T is a *spanning tree* of G if:

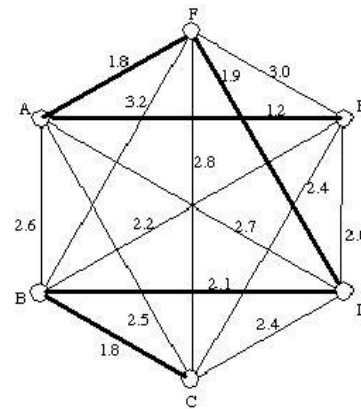
- T is a subgraph of G
- and
- T contains all the vertices of G



Minimal spanning trees

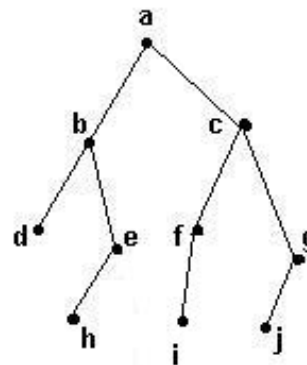
Given a weighted graph G , a *minimum spanning tree* is

- ▣ a spanning tree of G
- ▣ that has minimum “weight”

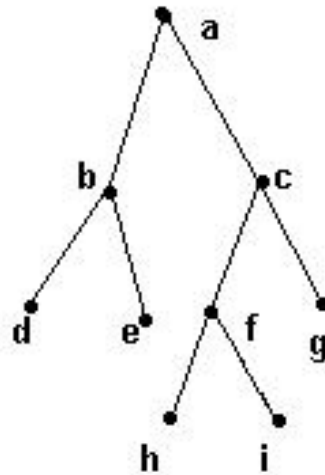


Binary trees

A *binary tree* is a tree where each vertex has zero, one or two children



Full binary tree



A *full* binary tree is a binary tree in which each vertex has two or no children.